

Final Examination

Course: Artificial Intelligence

AI-Assisted Machine Learning Project

Time: 3 Hours

Total Marks: 100

Instructions

- AI tools (ChatGPT, Gemini, Claude, Copilot, etc.) are allowed.
- Use Python, Google Colab/Jupyter Notebook and online documentation.
- Justify every important decision.
- Marks are based on understanding, critical thinking and explanation.
- AI-generated code without explanation will receive low marks.

Scenario

You are hired as a Junior AI Engineer by HealthPlus Hospital. Build an AI system to predict heart disease using the provided dataset.

Task 1: Dataset Understanding (10 Marks)

- Identify target variable.
- Classification or Regression?
- Identify numerical and categorical features.
- Suggest important features.
- Mention two challenges.
- Explain how AI can assist.

Task 2: Exploratory Data Analysis (15 Marks)

- Dataset dimensions, data types, missing values, duplicates, statistical summary, class distribution, correlation matrix.
- Create at least six meaningful visualizations.
- Interpret every visualization.

Task 3: Data Cleaning & Preprocessing (15 Marks)

- Handle missing values, duplicates, outliers.
- Encode categorical variables.
- Scale features if necessary.
- Explain every preprocessing decision and alternatives.

Task 4: Feature Engineering (15 Marks)

- Create meaningful new features.

- Perform transformations if appropriate.
- Select/remove features.
- Explain why features were created or removed.
- Compare performance before and after feature engineering.

Task 5: Machine Learning Models (20 Marks)

- Build at least three ML models.
- Compare results.
- Select the best model and justify.

Task 6: Model Evaluation (10 Marks)

- Report Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix.
- Explain the most appropriate metric.

Task 7: Model Interpretation (5 Marks)

- Show Feature Importance (or SHAP/LIME).
- Explain the top 10 important features.

Task 8: AI-Assisted Development Reflection (5 Marks)

- Include AI prompts, AI outputs, your modifications, AI mistakes, and improvements.

Task 9: Critical Discussion & Recommendations (5 Marks)

- Discuss limitations, bias, data leakage, ethics, privacy, deployment recommendation.

Submission

- Jupyter Notebook (.ipynb)
- Python Script (.py)
- Final Report (PDF)
- AI Prompt History
- Cleaned Dataset (CSV)

Marking Rubric

Component	Marks
Dataset Understanding	10
EDA	15
Data Cleaning	15
Feature Engineering	15
ML Models	20

Model Evaluation	10
Model Interpretation	5
AI Reflection	5
Critical Discussion	5
Total	100